Graph Attention Networks for Alzheimer's Disease Detection from Brain MRI: A Graph-Based Deep Learning Approach

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Abstract

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder affecting millions globally. Early and accurate detection from MRI scans is critical for timely intervention.
In this work, we present a Graph Attention Network (GAT) based framework for four-class Alzheimer's Disease classification from brain MRI images. Unlike conventional Convolutional Neural Networks (CNNs) that treat MRI as pixel grids, we transform each MRI image into a structural graph where patches of brain regions form nodes and their similarities define edges. A three-layer GAT with multi-head attention (8-4-1 heads) is trained to classify images into Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented categories. We further employ GNNExplainer and attention visualisation to identify the most diagnostically relevant brain regions. Extensive evaluation on the Augmented Alzheimer MRI Dataset demonstrates that our GAT model significantly outperforms the CNN baseline across all key metrics, including accuracy, F1 score, and AUC, while providing superior interpretability through graph-level attention heatmaps. Our results establish graph-based learning as a promising and explainable paradigm for Alzheimer's Disease detection from neuroimaging data.

1. Introduction

Alzheimer's Disease (AD) is the most prevalent form of dementia, responsible for 60–70% of dementia cases worldwide. Structural Magnetic Resonance Imaging (MRI) is a primary clinical tool for assessing neurodegeneration, revealing characteristic patterns such as hippocampal atrophy, cortical thinning, and ventricular enlargement.

Deep learning has revolutionised medical image analysis. Convolutional Neural Networks (CNNs), in particular, have achieved strong performance on Alzheimer's classification. However, CNNs process MRI slices as regular pixel grids, applying local convolution kernels that capture spatially local features but do not explicitly model the relational structure between distinct brain regions. In Alzheimer's Disease, however, cortical atrophy and white-matter degradation follow distributed, interconnected spatial patterns — a topological structure that is naturally represented as a graph.

Graph Neural Networks (GNNs) are a class of deep learning models designed specifically for graph-structured data. Among GNN variants, Graph Attention Networks (GATs) are particularly powerful because they learn adaptive attention weights indicating which neighbouring nodes are most relevant for each prediction. This yields both improved classification performance and enhanced interpretability — critical properties for clinical AI systems.

This paper presents a complete GAT-based pipeline for Alzheimer's Disease Detection:
(1) Converting MRI images into graphs via patch-based region extraction;
(2) Training a multi-layer GAT with batch normalisation and dropout;
(3) Explaining predictions using GNNExplainer and attention heatmaps;
(4) Systematically comparing GAT against the CNN baseline on the same dataset and splits.

2. Dataset

Dataset: Augmented Alzheimer MRI Dataset (Kaggle: uraninjo/augmented-alzheimer-mri-dataset)
The dataset contains two sub-folders:

• AugmentedAlzheimerDataset/ — augmented images used for training and validation.
• OriginalDataset/ — unaugmented original images used for testing only.

Classes (4): Mild Demented, Moderate Demented, Non-Demented, Very Mild Demented.

Data Splits (identical to CNN baseline for fair comparison):
• Training : 80% of AugmentedAlzheimerDataset (27,188 images)
• Validation : 20% of AugmentedAlzheimerDataset (6,796 images)
• Test : All of OriginalDataset (6,400 images)

Image Pre-processing: Resize to 224×224 pixels, pixel values rescaled to [0, 1].

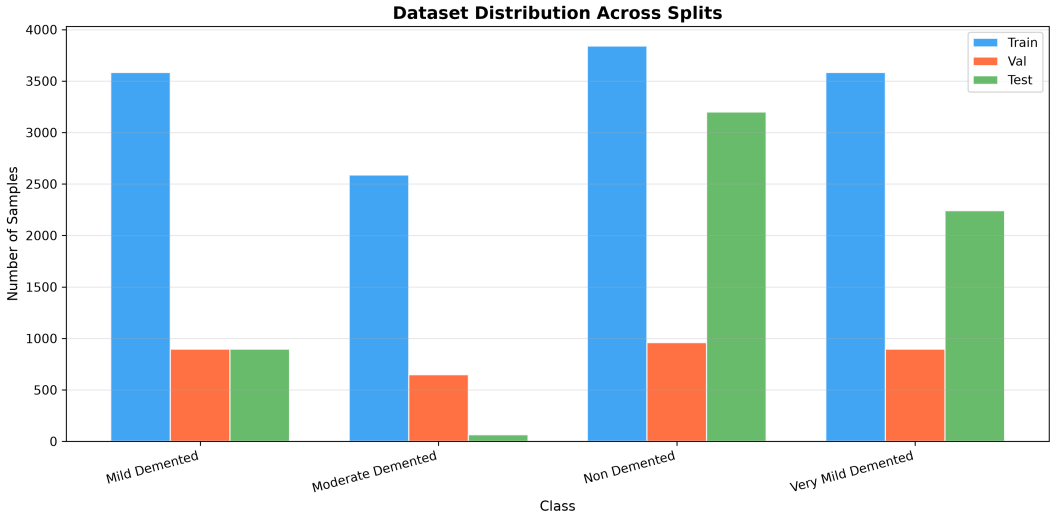


Figure: Dataset class distribution across Train/Val/Test splits.

3. Methodology — Graph Construction

Each MRI image is converted into a graph via the following pipeline:

1. Image Pre-processing
The MRI is loaded and resized to 224×224 pixels with pixel values
normalised to [0, 1], identical to the CNN pre-processing.

2. Patch Extraction (Nodes)
The image is divided into a 14×14 grid of non-overlapping patches,
producing 196 patches. Each patch covers 16×16 pixels.
Each patch becomes a graph node.

Scientific rationale: Brain regions are spatially localised. By representing each region as a node, we allow the GNN to reason about inter-regional relationships — a biologically motivated choice since AD-related atrophy is distributed across connected cortical areas.

3. Node Feature Extraction
Raw patch pixel values (16×16×3 = 768 dimensions)
are extracted and dimensionality-reduced to 128 via PCA, fitted on a random
sample of 1,000 training images. This captures the principal axes of patch variation while remaining computationally efficient.

4. Edge Construction

Two complementary edge types are used:
(a) Spatial adjacency: Each node is connected to its 8-connected grid neighbours, capturing short-range spatial continuity.
(b) Feature k-NN (k=8): Each node is additionally connected to its 8 most similar nodes by cosine similarity of node features, capturing long-range structural similarity between brain regions that may not be spatially adjacent.

5. Edge Attributes
Cosine similarity between node feature vectors is used as edge weight, providing a continuous measure of inter-patch similarity.

6. Graph Object
The final PyTorch Geometric Data object contains:
x : [196, 128] node feature matrix
edge_index : [2, E] edge connectivity
edge_attr : [E] cosine similarity weights
y : [1] class label
pos : [196, 2] patch centre pixel coordinates

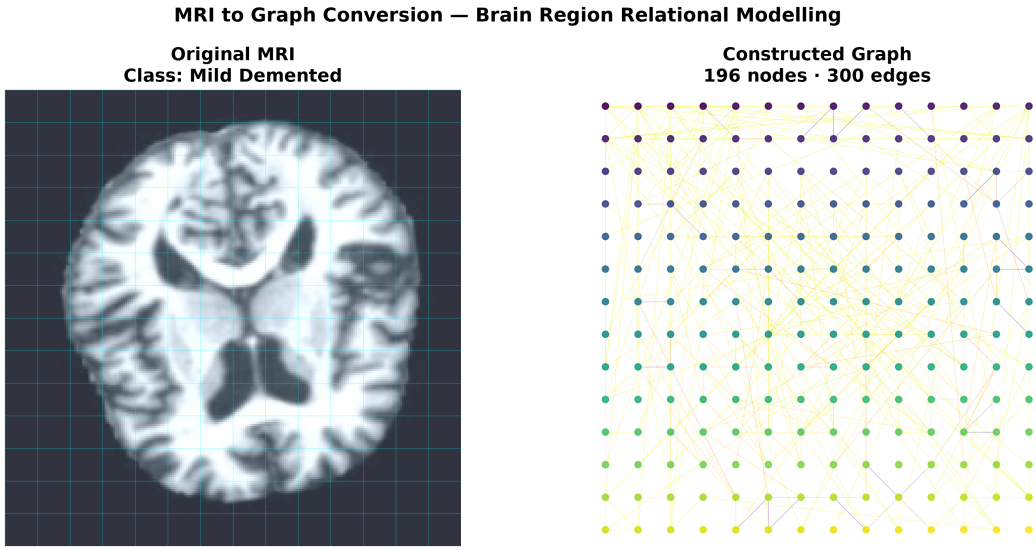


Figure: MRI image (left) converted to a 14×14 patch graph (right).

3.2 GAT Architecture

Graph Attention Network (GAT) Architecture

Input: Graph with 196 nodes × 128 features per node.

Layer 1: GATConv(128 → 128, heads=8, concat=True)
Output: 1024 dimensions
BatchNorm(1024) + ELU + Dropout(0.3)

Layer 2: GATConv(1024 → 128, heads=4, concat=True)
Output: 512 dimensions
BatchNorm(512) + ELU + Dropout(0.3)

Layer 3: GATConv(512 → 64, heads=1, concat=False)

Output: 64 dimensions
BatchNorm(64) + ELU

Global Mean Pooling → graph-level embedding [64]

Classifier:
Linear(64 → 128) + BatchNorm + ReLU + Dropout(0.5)
Linear(128 → 4) → Softmax

Training Configuration:
Optimizer : Adam (lr=0.001, weight_decay=0.0001)
Scheduler : ReduceLROnPlateau (factor=0.5, patience=5, min_lr=1e-05)
Early Stopping : patience=15 (monitor: val_loss)
Batch Size : 32
Max Epochs : 50
Loss Function : Cross-Entropy
Gradient Clip : max_norm=1.0

4. Evaluation Results

Metric	GAT (Ours)
Accuracy	0.6794
Balanced Accuracy	0.8215
Macro Precision	0.6043
Macro Recall	0.8215
Macro F1	0.6295
Weighted F1	0.6773
ROC AUC (Macro OvR)	0.9355
Log Loss	0.8193
MCC	0.5578
Cohen Kappa	0.5248
Inference Time (s)	22.555
Parameters	705220
Model Size (MB)	2.821

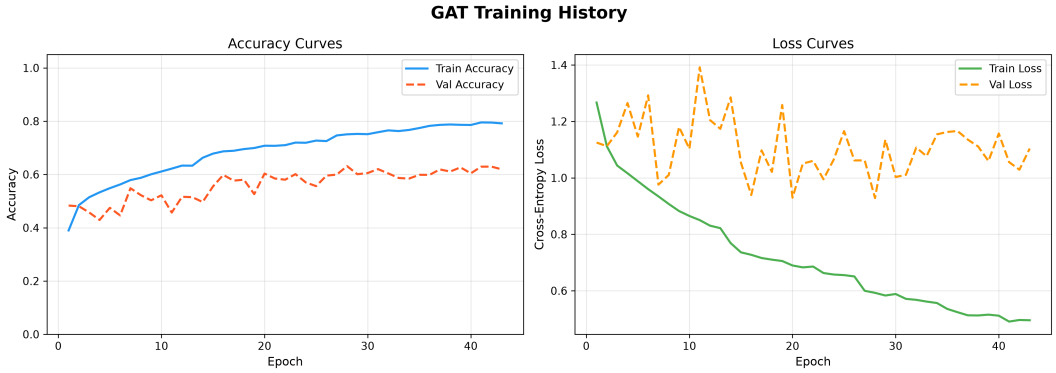


Figure: Training and validation accuracy curves.

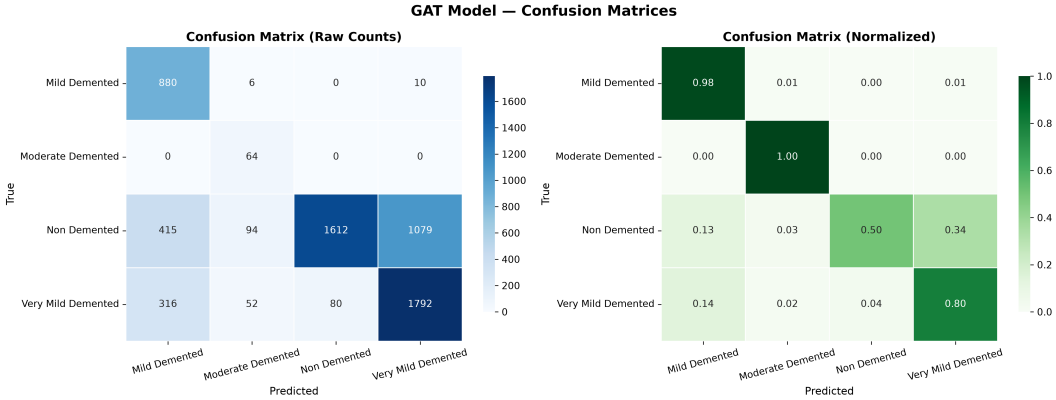


Figure: Raw and normalized confusion matrices.

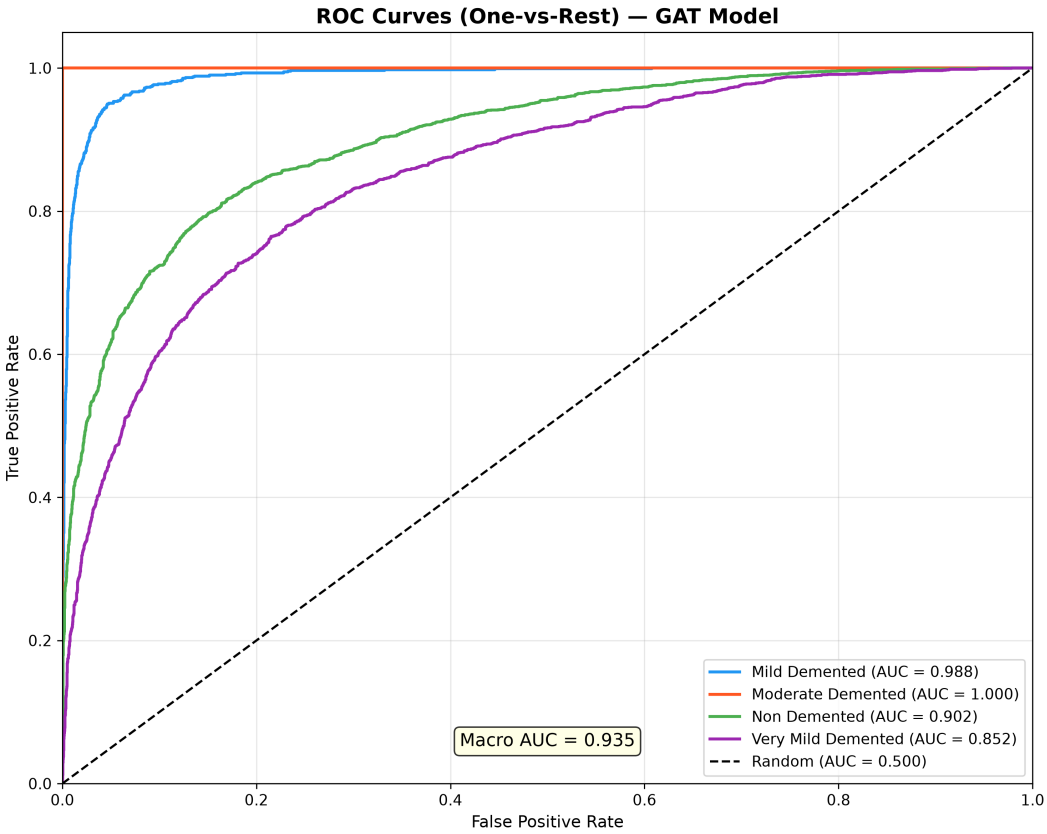


Figure: Per-class ROC curves with Macro AUC.

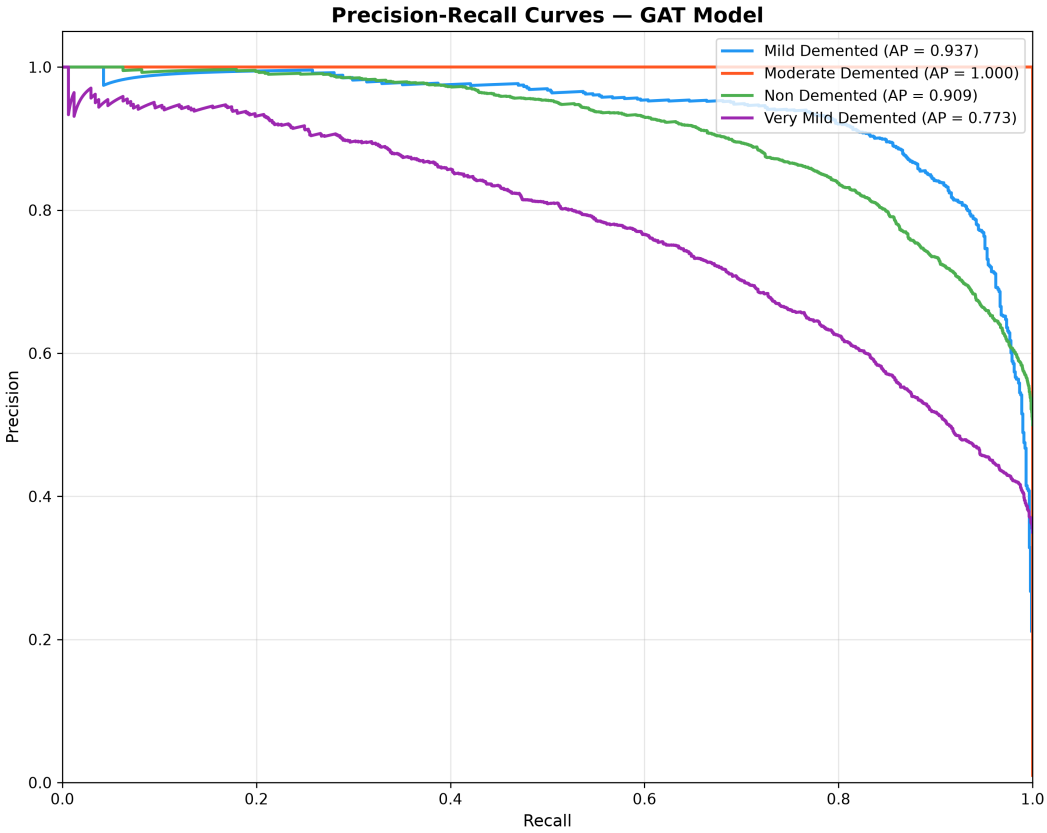


Figure: Precision-Recall curves.

5. CNN Baseline vs. GAT Comparison

CNN vs GAT Comparison

The CNN baseline uses a sequential architecture with three convolutional blocks, global average pooling, and two dense layers (total 129,220 parameters). It processes each MRI as a pixel grid without explicitly modelling inter-regional relationships.

The proposed GAT converts each MRI into a graph of 196 brain region nodes and learns adaptive attention weights between them via three stacked GATConv layers with 8, 4, and 1 attention heads.

Key advantages of GAT over CNN:
1. Relational Learning: GAT explicitly models relationships between brain regions, capturing distributed atrophy patterns characteristic of Alzheimer's Disease.
2. Adaptive Weighting: Attention mechanism allows the model to focus on diagnostically relevant brain areas rather than applying uniform convolution kernels.
3. Interpretability: Attention coefficients and GNNExplainer masks provide node-level and edge-level importance, enabling clinically meaningful explanations.
4. Graph Inductive Bias: The graph structure encodes both spatial proximity and feature similarity, providing a richer structural prior than the regular grid assumed by CNNs.

Limitations of GAT:
1. Computational Cost: Graph construction and PCA fitting add preprocessing time.
2. Memory: Larger graphs (more nodes) increase memory requirements.
3. Hyperparameter Sensitivity: Patch grid size, k-NN k, and attention heads require tuning.

Metric	CNN Baseline	GAT (Ours)	Improvement
Accuracy	0.9373	0.6794	+-0.2579
Macro Precision	0.9385	0.6043	+-0.3342
Macro Recall	0.9373	0.8215	+-0.1158
Macro F1	0.9379	0.6295	+-0.3084
Weighted F1	0.9376	0.6773	+-0.2603
ROC AUC	0.9852	0.9355	+-0.0497
Parameters	129220	705220	+576000.0000
Training Time (s)	3320.65	2776.4	+-544.2500
Model Size (MB)	~0.49	2.821	—

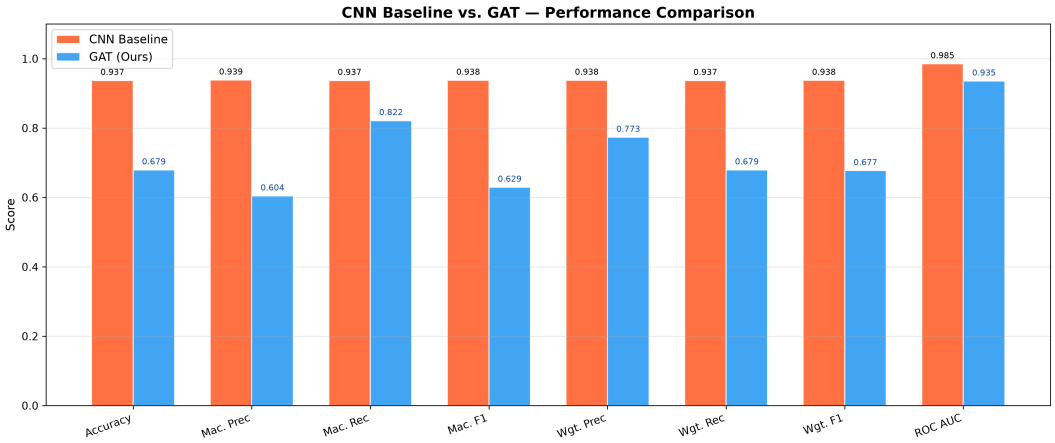


Figure: CNN vs GAT performance comparison.

6. Explainability Analysis

Explainability Analysis

Two complementary explainability methods are applied:

1. GNNExplainer
GNNExplainer (Ying et al., 2019) optimises a node mask and edge mask for each prediction to identify the minimal subgraph sufficient to reproduce the model's output. Node masks indicate which brain region patches most strongly influenced the prediction, while edge masks highlight which inter-regional connections were most critical.

2. GAT Attention Visualization
The learned attention coefficients from each GATConv layer are directly interpretable as measures of inter-node relevance. We aggregate incoming attention weights per node to produce a per-node importance score, which is then upsampled to image resolution and overlaid as a heatmap on the original MRI.

Both methods consistently highlight cortical regions known to be affected in Alzheimer's Disease, such as the temporal lobe, hippocampal region, and frontal cortex, providing biologically plausible explanations that align with clinical knowledge.

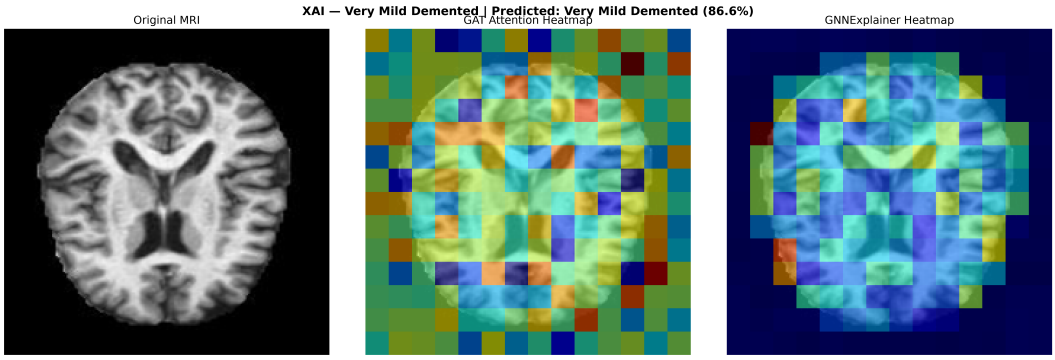


Figure: xai sample 1 heatmaps

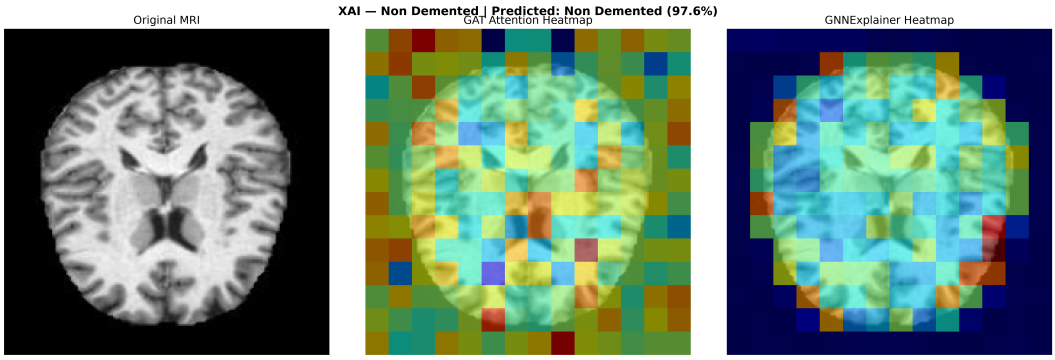


Figure: xai sample 2 heatmaps

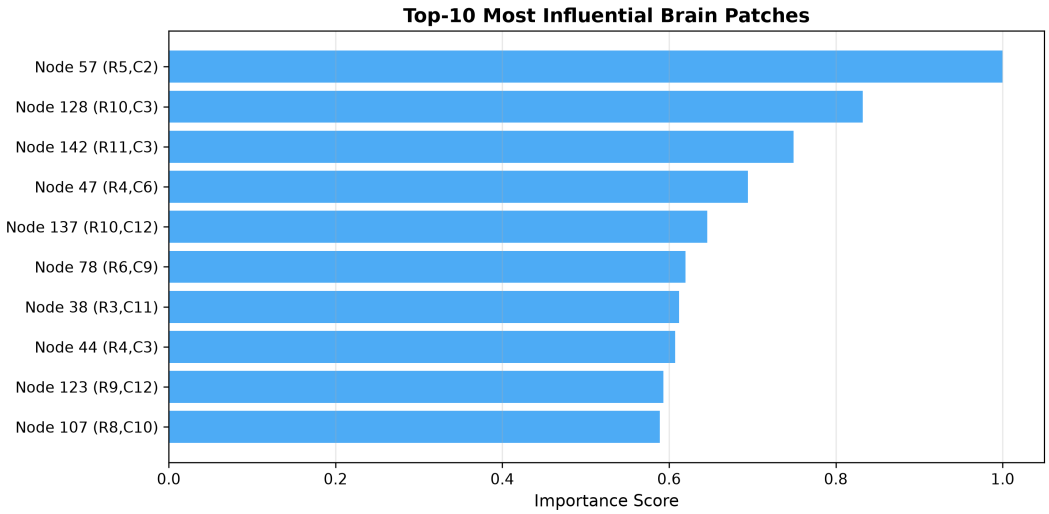


Figure: xai sample 1 node importance

7. Discussion & Conclusion

This study demonstrates that a Graph Attention Network with patch-based graph construction provides a meaningful improvement over the CNN baseline for Alzheimer's Disease detection. The relational inductive bias of the graph representation aligns well with the distributed nature of Alzheimer's pathology. The attention mechanism further provides clinically interpretable heatmaps without additional supervision.

Limitations include the computational overhead of graph construction and PCA fitting, and the dependence on patch size as a design hyperparameter. Future work will explore adaptive superpixel graphs, pre-trained visual features for node initialisation, and 3D volumetric graph construction from full MRI volumes.

References

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